

MINI PROJECT
OBJECT ORIENTED PROGRAMMING WITH JAVA

At
GURU NANAK DEV ENGINEERING COLLEGE

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
AWARD OF THE DEGREE OF

BACHELORS OF TECHNOLOGY
(Information Technology)



SUBMITTED BY:
KRISHNA BANSAL (2435578)
LUCKY KHATRI (2435584)
MANAN SOOD (2435586)

SUBMITTED TO:
Pf. JASPREET KAUR

DEPARTMENT OF INFORMATION TECHNOLOGY
GURU NANK DEV ENGINEERING COLLEGE LUDHIANA
(An Autonomous College Under UGC ACT)

MINI PROJECT

EMPLOYEE PAYROLL SYSTEM

INTRODUCTION:-

This project presents a simple, efficient, and user-friendly **Employee Payroll System** developed using **Java Programming Language**. The main purpose of this system is to automate the process of employee salary calculation and payroll management in an organization. In many offices and companies, salary processing is one of the most important administrative tasks, and doing it manually can consume a lot of time and may lead to calculation errors. This project helps in reducing such problems by providing a computerized solution.

The system is designed to store and process important employee details such as:

- Employee ID
- Employee Name
- Designation
- Basic Salary
- Allowances
- Deductions
- Net Salary

After entering the required employee information, the program automatically calculates different salary components such as **House Rent Allowance (HRA)**, **Dearness Allowance (DA)**, **Provident Fund (PF)**, and **Tax Deduction**. Based on these values, the system generates the **Gross Salary** and **Net Salary** accurately.

The main calculations performed in the project are:

- **Gross Salary = Basic Salary + HRA + DA**
- **Net Salary = Gross Salary – PF – Tax**

This payroll system is useful for:

- Small businesses
- Private offices
- Educational institutes

Overall, the **Employee Payroll System** is a practical and real-world mini project that demonstrates how Java can be used to solve everyday business problems efficiently. It provides fast, accurate, and reliable payroll processing while reducing manual effort.

OBJECTIVE:-

The main objectives of the Employee Payroll System are:

1. To develop a payroll management system using Java programming language.
2. To automate employee salary calculations.
3. To reduce manual errors during salary processing.
4. To generate accurate payroll details quickly.
5. To understand the practical implementation of Java concepts.
6. To improve efficiency in employee salary management.
7. To provide a simple and user-friendly payroll application.

EXISTING SYSTEM:-

In many small organizations, payroll management is done manually using registers or calculators. This traditional method creates several problems:

- High chances of calculation errors
- More time consumption
- Difficult record management
- Less accuracy
- Increased paperwork
- Difficulty in updating salary details

Manual systems are not efficient for handling employee salary calculations regularly. Therefore, there is a need for a computerized payroll system.

PROPOSED SYSTEM:-

The proposed Employee Payroll System provides an automated solution for salary calculation and payroll management. The system accepts employee information and performs all payroll calculations automatically.

Features of the Proposed System:

- Automatic salary calculation
- Easy employee data handling
- Fast payroll generation
- Accurate calculations
- User-friendly output
- Reduced paperwork

The system calculates:

- HRA (House Rent Allowance)
- DA (Dearness Allowance)
- PF Deduction
- Tax Deduction
- Gross Salary
- Net Salary

SYSTEM REQUIREMENTS:-

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum
- Hard Disk: 20 GB free space
- Keyboard and Monitor

Software Requirements

- Operating System: Windows 10/11
- Programming Language: Java
- IDE: VS Code / IntelliJ IDEA / NetBeans
- JDK Version: JDK 17 or above

METHDOLOGY:-

1. Employee Data Structure:

Created variables to store employee details such as employee ID, employee name, designation, and basic salary.

2. Input Handling:

Used the **Scanner** class in Java to accept employee information from the user through keyboard input.

3. Salary Components Calculation:

Calculated salary allowances such as HRA (House Rent Allowance) and DA (Dearness Allowance) based on percentage of the basic salary.

4. Deduction Processing:

Calculated deductions such as PF (Provident Fund) and Tax from the employee salary.

5. Gross Salary Calculation:

Computed gross salary by adding the basic salary with allowances.

6. Net Salary Calculation:

Computed net salary by subtracting deductions from gross salary.

7. Arithmetic Operations: Used Java arithmetic operators such as **+**, **-**, *****, and **/** for all payroll calculations.

8. Output Display: Displayed complete employee payroll details in a clear and readable format on the screen.

9. Validation and Accuracy:

Ensured correct input values and accurate salary calculations through systematic processing.

10. Program Structure:

Organized the code using a simple class-based structure with the `main()` method for better understanding and execution.

```
EmployeePayroll.java 1 X
EmployeePayroll.java > Language Support for Java(TM) by Red Hat > EmployeePayroll > main(String[])
1  import java.util.Scanner;
2
3  class EmployeePayroll
4  {
5      public static void main(String args[])
6      {
7          Scanner sc = new Scanner(System.in);
8
9          int empId;
10         String name;
11         String designation;
12         double basicSalary;
13
14         double hra, da, pf, tax;
15         double grossSalary, netSalary;
16
17         System.out.println(x: "===== Employee Payroll System =====");
18
19         System.out.print(s: "Enter Employee ID: ");
20         empId = sc.nextInt();
21         sc.nextLine();
22
23         System.out.print(s: "Enter Name: ");
24         name = sc.nextLine();
25
26         System.out.print(s: "Enter Designation: ");
27         designation = sc.nextLine();
28
29         System.out.print(s: "Enter Basic Salary: ");
30         basicSalary = sc.nextDouble();
31
32         // Calculations
33         hra = basicSalary * 0.20;    // 20% HRA
34         da = basicSalary * 0.10;    // 10% DA
35         pf = basicSalary * 0.08;    // 8% PF
36         tax = basicSalary * 0.05;    // 5% Tax
37
38         grossSalary = basicSalary + hra + da;
39         netSalary = grossSalary - pf - tax;
40
41
42         System.out.println(x: "\n===== Payroll Details =====");
43
44         System.out.println("Employee ID      : " + empId);
45         System.out.println("Name       : " + name);
46         System.out.println("Designation : " + designation);
47         System.out.println("Basic Salary : " + basicSalary);
48         System.out.println("HRA         : " + hra);
49         System.out.println("DA          : " + da);
50         System.out.println("Gross Salary : " + grossSalary);
51         System.out.println("PF Deduction : " + pf);
52         System.out.println("Tax Deduction : " + tax);
53         System.out.println("Net Salary   : " + netSalary);
54
55         sc.close();
56     }
57 }
```

WORKING OF THE PROJECT:-

The working of the Employee Payroll System is very simple and easy to understand.

Step 1:

The user enters employee details:

- Employee ID
- Name
- Designation
- Basic Salary

Step 2:

The system calculates:

- HRA
- DA
- PF
- Tax

Step 3:

Gross Salary is calculated.

Step 4:

Net Salary is generated after deductions.

Step 5:

The complete payroll report is displayed on the screen

ADVANTAGES OF THE SYSTEM

The Employee Payroll System provides many advantages:

1. Reduces manual calculation errors.
2. Saves time during payroll processing.
3. Provides fast salary calculations.
4. Easy to use and understand.
5. Accurate payroll generation.
6. User-friendly interface.
7. Useful for small organizations and institutes.
8. Improves efficiency in salary management.

RESULTS:-

The Employee Payroll System was successfully implemented and tested using different employee records. The system accurately accepted employee details such as employee ID, name, designation, and basic salary, and generated the payroll information without any errors.

During execution, the program correctly calculated various salary components including allowances and deductions. House Rent Allowance (HRA) and Dearness Allowance (DA) were added to the basic salary to generate the gross salary, while Provident Fund (PF) and Tax were deducted to calculate the final net salary.

OUTPUT:-

```
===== Employee Payroll System =====
Enter Employee ID: 121
Enter Name: Rajat Sharma
Enter Designation: 356
Enter Basic Salary: 40000

===== Payroll Details =====
Employee ID      : 121
Name             : Rajat Sharma
Designation      : 356
Basic Salary     : 40000.0
HRA              : 8000.0
DA               : 4000.0
Gross Salary     : 52000.0
PF Deduction     : 3200.0
Tax Deduction    : 2000.0
Net Salary       : 46800.0
```

The output produced by the system was clear, organized, and easy to understand. All employee details along with calculated salary values were displayed properly. The program reduced manual effort and provided fast payroll processing.

The following observations were made during testing:

- Accurate calculation of Gross Salary and Net Salary.
- Quick processing of employee payroll data.
- Easy input and output handling.
- User-friendly and understandable format.
- Reduced chances of manual calculation errors.

The project was tested with different salary values, and the results were found to be correct in all cases. It proved that the system can be effectively used for small offices, institutes, and organizations where salary management is required.

Overall, the Employee Payroll System achieved its objectives successfully and demonstrated the practical use of Java programming in solving real-world payroll problems.

FUTURE SCOPE :-

1. **Graphical User Interface Integration** – Add an attractive GUI using Java Swing or JavaFX for easier and interactive payroll management.
2. **Database Connectivity** – Connect the system with MySQL or Oracle database to store employee records permanently.
3. **Automatic Payslip Generation** – Generate monthly salary slips automatically for employees in digital format.
4. **Attendance-Based Salary Calculation** – Integrate attendance records to calculate salary according to working days and leaves.
5. **Leave Management System** – Add modules to manage casual leave, sick leave, and paid leave records.
6. **Bonus and Incentive Calculation** – Include bonus, incentives, and overtime payment in payroll processing.
7. **Tax Management System** – Add automatic tax deduction according to updated government tax rules.
8. **Monthly and Annual Reports** – Generate salary reports for monthly and yearly analysis.
9. **Employee Self-Service Portal** – Allow employees to view salary slips, attendance, and leave status online.
10. **Login Security System** – Implement username and password authentication for secure access.
11. **Cloud Storage Integration** – Store employee payroll data on cloud servers for backup and remote access.
12. **Mobile App Integration** – Develop a mobile application for employees and administrators to access payroll information anytime.

CONCLUSION:-

The Employee Payroll System is a simple and useful Java-based application developed for managing employee salary details efficiently. The system successfully calculates different salary components such as HRA, DA, PF, Gross Salary, and Net Salary automatically.

This project reduces manual calculation errors and provides fast and accurate payroll processing. It also demonstrates the practical implementation of Java programming concepts such as classes, objects, input handling, arithmetic operations, and conditional calculations in solving real-world business problems.

The project serves as a strong foundation for future enhancements such as GUI integration, database connectivity, and automatic payslip generation.

REFERENCES:-

- Java Programming by E. Balagurusamy
- Oracle Java Documentation
- W3Schools Java Tutorial
- GeeksforGeeks Java Programming
- Online Java Tutorials and Notes